

Chapter 8. Plasticity and Learning

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June 2, 2023

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Objective : Understand how experience and training modify synapses and how these modifications change patterns of neuronal firing

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8.1 Introduction

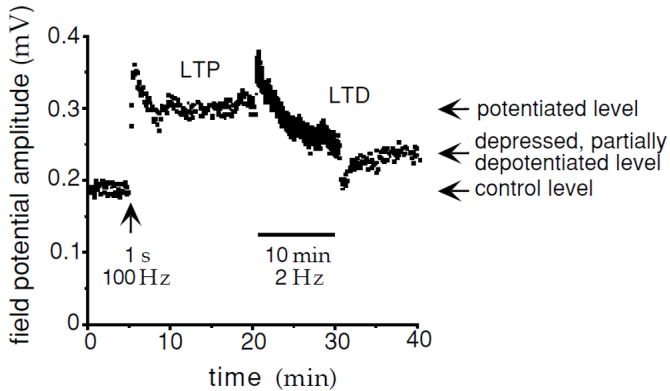
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Introduction to Plasticity

- ([Hebb rule](#)) If input from neuron A contributes to the firing of neuron B, then the synapse from A to B should be strengthened.
- ([LTP and LTD](#)) Changes that persist for tens of minutes or longer are called long-term potentiation and long-term depression.
- ([Unsupervised Learning](#)) A training procedure in which a network self-organizes itself in a manner that depends on the plasticity rule and the nature of inputs.
- ([Supervised Learning](#)) A training procedure in which a desired set of (input, output) relations are imposed on the network.



Stability and Competition

- (**Synaptic Saturation**) Imposing an upper limit on the value of a synaptic weight controls synaptic strengthening.
- (**Synaptic Competition**) Input selectivity requires some synapses to weaken when others become strong. (If all synapses are saturated, the network loses selectivity to patterns)

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The Basic Hebb Rule

- Suppose u (presynaptic activity), v (postsynaptic activity) are dimensionless measures in $[0, \infty)$
- The basic Hebb rule is given by (8.3). It is **unstable and noncompetitive**.

$$\tau_r \frac{dv}{dt} = -v + \mathbf{w} \cdot \mathbf{u} = -v + \sum_{b=1}^{N_u} w_b u_b, \quad (8.1)$$

$$v = \mathbf{w} \cdot \mathbf{u}, \quad (8.2)$$

$$\tau_w \frac{d\mathbf{w}}{dt} = v \mathbf{u}, \quad (8.3)$$

The Averaged Hebb Rule

- The averaged Hebb rule takes the average of the RHS, which gives us

$$\tau_w \frac{d\mathbf{w}}{dt} = \langle v\mathbf{u} \rangle. \quad (8.4)$$

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w} \quad \text{or} \quad \tau_w \frac{dw_b}{dt} = \sum_{b'=1}^{N_u} Q_{bb'} w_{b'}, \quad (8.5)$$

$$\mathbf{Q} = \langle \mathbf{u}\mathbf{u} \rangle \quad \text{or} \quad Q_{bb'} = \langle u_b u_{b'} \rangle. \quad (8.6)$$

The Covariance Rule

- We need to impose a saturation constraint on the basic Hebb rule.
- postsynaptic threshold θ_v and presynaptic threshold θ_u model LTD as well as LTP by the following rules:

$$\tau_w \frac{d\mathbf{w}}{dt} = (v - \theta_v) \mathbf{u}, \quad (8.8)$$

$$\tau_w \frac{d\mathbf{w}}{dt} = v(\mathbf{u} - \theta_u). \quad (8.9)$$

- Suppose we chose $\theta_v = \langle v \rangle$, $\theta_u = \langle u \rangle$. Then the heterosynaptic depression model becomes (8.10) with the covariance matrix $\mathbf{C}$.
- However, this "covariance" model is still **unstable**! Competition can be introduced by allowing thresholds to "slide".

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{C} \cdot \mathbf{w}, \quad (8.10)$$

$$\mathbf{C} = \langle (\mathbf{u} - \langle \mathbf{u} \rangle)(\mathbf{u} - \langle \mathbf{u} \rangle) \rangle = \langle \mathbf{u}\mathbf{u} \rangle - \langle \mathbf{u} \rangle^2 = \langle (\mathbf{u} - \langle \mathbf{u} \rangle)\mathbf{u} \rangle. \quad (8.11)$$

The BCM Rule

- Bienenstock, Cooper, and Munro proposed the BCM rule in 1982. One instantiation of the BCM rule is with a [sliding threshold](#).
- Let the postsynaptic threshold θ_v "slide" as in (8.13). Then strengthening some synapses increases v , which raises θ_v , and makes it more difficult for other synapses to be strengthened.
(i.e., We have [competition](#) between synapses)

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u}(v - \theta_v) . \quad (8.12)$$

$$\tau_\theta \frac{d\theta_v}{dt} = v^2 - \theta_v . \quad (8.13)$$

Synaptic (weight) Normalization

- Normalization can drastically alter the outcome of a training procedure.
- **Global constraints** enforce $\sum w_i = \text{constant}$ if $w_i \geq 0$ or $\sum w_i^2 = \text{constant}$ if $w_i \in \mathbb{R}$
- **Rigid** constraints are always satisfied, while **dynamic** constraints are asymptotically satisfied.
- We discuss one example of each type.

Subtractive Normalization

- This rule imposes the constraint on the sum of weights rigidly because it does not allow the Hebbian rule to change $\mathbf{n} \cdot \mathbf{w} = \sum w_i$. (8.15)
- Subtractive normalization must satisfy the precondition : $w_i \geq 0 \forall i$. This can be enforced by modifying $\mathbf{n}$ whenever one or more $w_i < 0$.
- This is highly **competitive**. Also, upper saturation limits are usually imposed to prevent the weight vector from becoming a one-hot vector.

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \frac{v(\mathbf{n} \cdot \mathbf{u})\mathbf{n}}{N_u}. \quad (8.14)$$

$$\tau_w \frac{d\mathbf{n} \cdot \mathbf{w}}{dt} = v\mathbf{n} \cdot \mathbf{u} \left(1 - \frac{\mathbf{n} \cdot \mathbf{n}}{N_u} \right) = 0. \quad (8.15)$$

Multiplicative Normalization

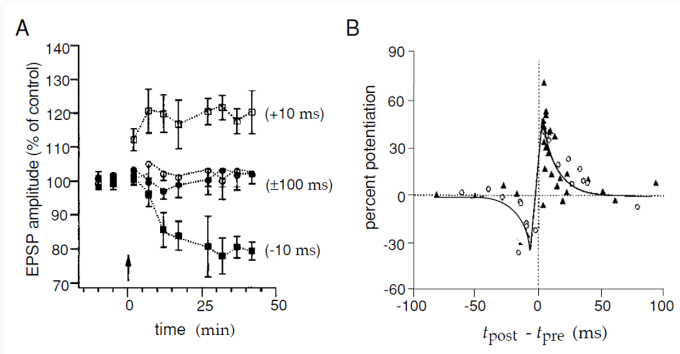
- A dynamic constraint on $\sum w_i^2$ can be imposed using the Oja rule (Oja, 1982).
- The rule only involves weights and activities local to each synapse being modified.
- The **stability and competitiveness** of this rule follows from (8.17), which indicates that $|\mathbf{w}|^2$ will relax over time to $1/\alpha$.

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \alpha v^2 \mathbf{w}, \quad (8.16)$$

$$\tau_w \frac{d|\mathbf{w}|^2}{dt} = 2v^2(1 - \alpha|\mathbf{w}|^2). \quad (8.17)$$

Timing-Based Rules

- Experiments with *Xenopus* tadpoles show that relative timing Δt is critical in determining the **sign (type)** and **amplitude** of synaptic modification.
- PRE $\rightarrow$ POST ($\Delta t > 0$) produce LTP and POST $\rightarrow$ PRE ($\Delta t < 0$) produce LTD. Synaptic plasticity is observed to occur at $|\Delta t| \approx 50\text{ms}$



- A function $H(\tau)$ determines the rate of synaptic modification ($\tau := \Delta t$). The rate of change is then given by (8.18)
- $H(\tau) > 0$ represents LTP, $H(\tau) < 0$ represents LTD.
- The timing rule generates **competition**, but still requires saturation constraints.

$$\tau_w \frac{d\mathbf{w}}{dt} = \int_0^\infty d\tau (H(\tau)v(t)\mathbf{u}(t-\tau) + H(-\tau)v(t-\tau)\mathbf{u}(t)) . \quad (8.18)$$

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Single Postsynaptic Neuron

- The network should uncover patterns in the input data by themselves.
- We express $\mathbf{w}$ as the linear combination of eigenvectors of $\mathbf{Q}$ to uncouple the equations.
- For correlation or covariance matrices, we can always order the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0$

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w} \quad \text{or} \quad \tau_w \frac{dw_b}{dt} = \sum_{b'=1}^{N_u} Q_{bb'} w_{b'}, \quad (8.5)$$

$$\mathbf{w}(t) = \sum_{\mu=1}^{N_u} c_{\mu}(t) \mathbf{e}_{\mu}, \quad (8.19)$$

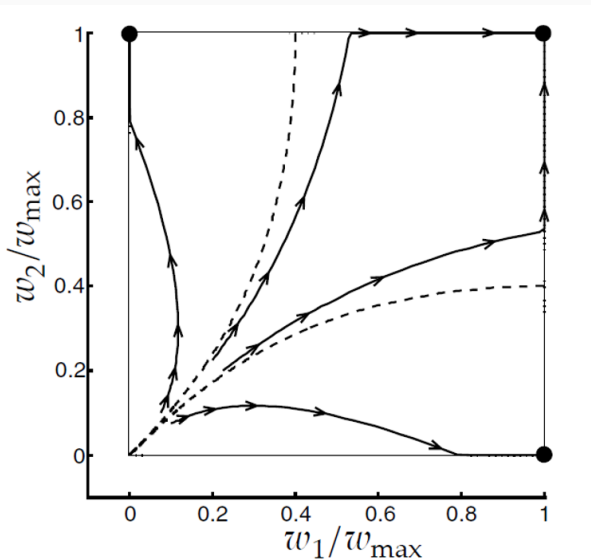
- For large t , plasticity produces output proportional to the projection of $\mathbf{u}$ onto the principal eigenvector $\mathbf{e}_1$

$$\mathbf{w}(t) = \sum_{\mu=1}^{N_u} \exp\left(\frac{\lambda_{\mu} t}{\tau_w}\right) (\mathbf{w}(0) \cdot \mathbf{e}_{\mu}) \mathbf{e}_{\mu} . \quad (8.20)$$

$$v \propto \mathbf{e}_1 \cdot \mathbf{u} . \quad (8.21)$$

How to control w

- Saturation constraint (simulation result)



- Oja rule: $\mathbf{w} \rightarrow \frac{1}{\sqrt{\alpha}} \mathbf{e}_1$
- Subtractive constraint : $\mathbf{w} \propto \mathbf{e}_2$

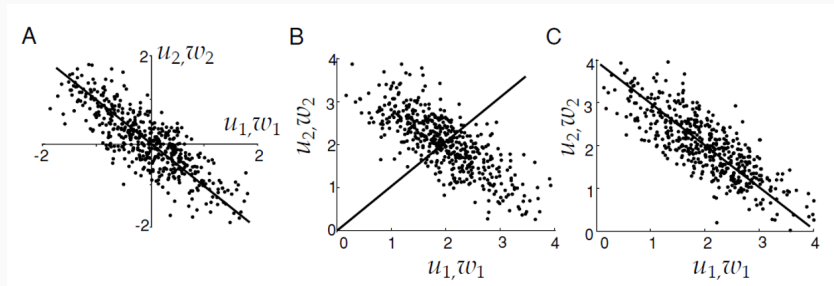
$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \frac{v(\mathbf{n} \cdot \mathbf{u})\mathbf{n}}{N_u} . \quad (8.14)$$

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w} - \frac{(\mathbf{w} \cdot \mathbf{Q} \cdot \mathbf{n})\mathbf{n}}{N_u} . \quad (8.22)$$

$$\mathbf{w}(t) = (\mathbf{w}(0) \cdot \mathbf{e}_1) \mathbf{e}_1 + \sum_{\mu=2}^{N_u} \exp\left(\frac{\lambda_{\mu} t}{\tau_w}\right) (\mathbf{w}(0) \cdot \mathbf{e}_{\mu}) \mathbf{e}_{\mu} . \quad (8.23)$$

Principal Component Projection

- $\mathbf{w} = \mathbf{e}_1$ is usually the best estimate which maximizes variance and output entropy.
- We will see more in Ch.10



Hebbian Development of Ocular Dominance

- Ocular dominance is the tendency to prefer visual input of one eye over the other.
- Since both eyes open and close, positive correlations $q_D > 0$ are likely to exist. Then w_+ becomes dominant over time and $\mathbf{w}$ becomes parallel with $\mathbf{e}_1$.

$$v = w_R u_R + w_L u_L. \quad (8.24)$$

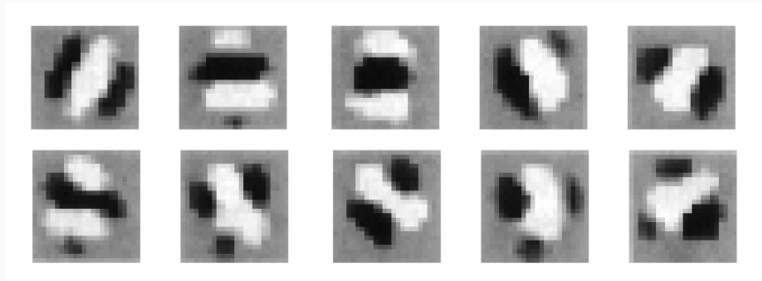
$$\mathbf{Q} = \langle \mathbf{u}\mathbf{u} \rangle = \begin{pmatrix} \langle u_R u_R \rangle & \langle u_R u_L \rangle \\ \langle u_L u_R \rangle & \langle u_L u_L \rangle \end{pmatrix} = \begin{pmatrix} q_S & q_D \\ q_D & q_S \end{pmatrix}. \quad (8.25)$$

$$\tau_w \frac{dw_+}{dt} = (q_S + q_D)w_+ \quad \text{and} \quad \tau_w \frac{dw_-}{dt} = (q_S - q_D)w_-. \quad (8.26)$$

- $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1)$ means there is no ocular dominance at all!
(Does not match observation)
- We solve this by applying Hebbian rule with subtractive constraint.
Then $\mathbf{w} \propto \mathbf{e}_2 = \frac{1}{\sqrt{2}}(1, -1)$
- Initial values determine growth or antigrowth or w_R . In any case, one dominates the other.

Hebbian Development of Orientation Selectivity

- Orientation selectivity relies on nonlinear aspects of the model and is difficult to study analytically. The book presents simulation results.



Temporal Hebbian Rules and Trace Learning

- Trace refers to the history of synaptic activity. We can approximate temporal plasticity rule (8.18) to obtain (8.27)
- Suppose an object moves across the retina for characteristic lengths of time. (8.27) strengthens synapses from all the presynaptic units that are activated by the image of the object while it persists and moves.
- The resulting response of the postsynaptic cell becomes independent of the position of the object.

$$\tau_w \frac{d\mathbf{w}}{dt} = \int_0^\infty d\tau (H(\tau)v(t)\mathbf{u}(t-\tau) + H(-\tau)v(t-\tau)\mathbf{u}(t)) . \quad (8.18)$$

$$\mathbf{w} = \frac{1}{\tau_w} \int_0^T dt v(t) \int_{-\infty}^\infty d\tau H(\tau)\mathbf{u}(t-\tau) . \quad (8.27)$$

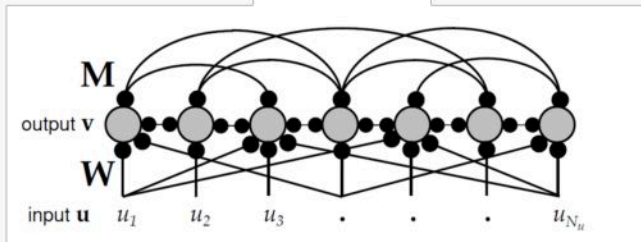
Multiple Postsynaptic Neurons

- Now we study the effect of plasticity on multiple neurons.

feedforward
weight matrix $\mathbf{W}$

$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}. \quad (8.28)$$

recurrent
weight matrix $\mathbf{M}$



- Fixed/plastic linear/nonlinear recurrent interactions can ensure that the units do not all develop the same selectivity.
- In the following sections, we consider 3 different cases: **W** is plastic, **M** is plastic, **W**, **M** are plastic

$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}. \quad (8.28)$$

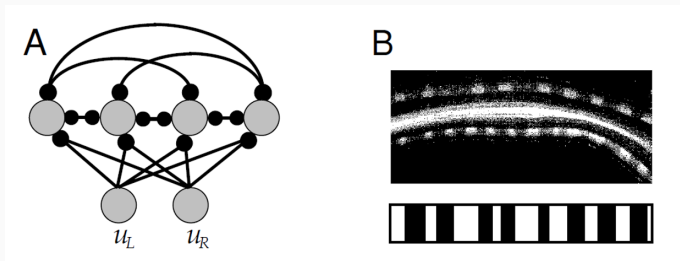
$$\mathbf{v} = \mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}. \quad (8.29)$$

$$\mathbf{v} = \mathbf{K} \cdot \mathbf{W} \cdot \mathbf{u} \quad \text{or} \quad v_a = \sum_{a'=1}^{N_v} \sum_{b=1}^{N_u} K_{aa'} W_{a'b} u_b. \quad (8.30)$$

$$\mathbf{K} = (\mathbf{I} - \mathbf{M})^{-1}$$

Hebbian Development of Ocular Dominance Stripes

- We use subtractive normalization, which fixes the growth of $\mathbf{w}_+$ while increasing $\mathbf{w}_-$
- "Stripes" are oscillations in the signs of $\mathbf{w}_-$
- We assume further that $\mathbf{K}_{ij} = K(|i - j|)$ and that the boundaries are periodic. Then we get the eigenvector for the a th output neuron as in (8.33).



$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w} \quad \text{or} \quad \tau_w \frac{dw_b}{dt} = \sum_{b'=1}^{N_u} Q_{bb'} w_{b'}, \quad (8.5)$$

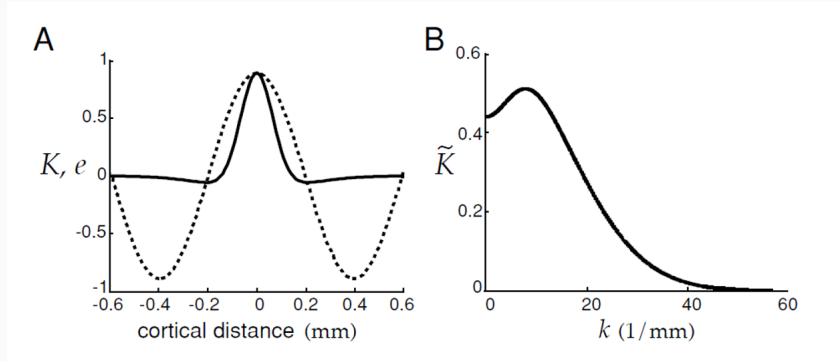
$$\tau_w \frac{d\mathbf{W}}{dt} = \langle \mathbf{v}\mathbf{u} \rangle = \mathbf{K} \cdot \mathbf{W} \cdot \mathbf{Q}, \quad (8.31)$$

$$\mathbf{Q} = \langle \mathbf{u}\mathbf{u} \rangle = \begin{pmatrix} \langle u_R u_R \rangle & \langle u_R u_L \rangle \\ \langle u_L u_R \rangle & \langle u_L u_L \rangle \end{pmatrix} = \begin{pmatrix} q_S & q_D \\ q_D & q_S \end{pmatrix}. \quad (8.25)$$

$$\tau_w \frac{d\mathbf{w}_+}{dt} = (q_S + q_D) \mathbf{K} \cdot \mathbf{w}_+ \quad \tau_w \frac{d\mathbf{w}_-}{dt} = (q_S - q_D) \mathbf{K} \cdot \mathbf{w}_-. \quad (8.32)$$

$$e_a^\mu = \cos \left(\frac{2\pi\mu a}{N_v} - \phi \right) \quad (8.33)$$

- A hypothetical K function and its Fourier transform.



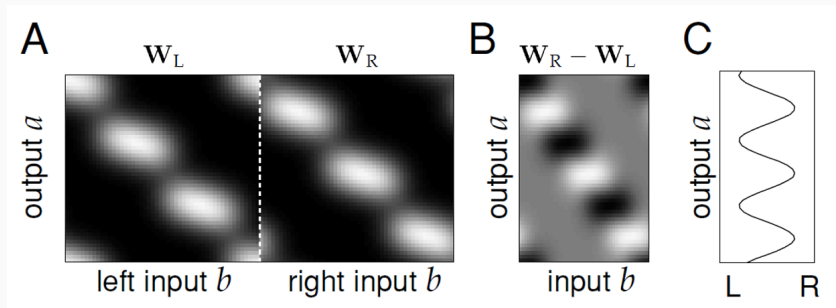
Competitive Hebb Rule

- [Stage 1: Long-range inhibition] Competition for feedforward input
- [Stage 2: Short-range excitation] Cooperation among neighbors
- Nonlinear interactions induce stronger competition in recurrent connections. Nonlinearity prevents the weight from becoming parallel to the principal eigenvector.

$$z_a = \frac{(\sum_b W_{ab} u_b)^\delta}{\sum_{a'} (\sum_b W_{a'b} u_b)^\delta} . \quad (8.34)$$

$$v_a = \sum_{a'} M_{aa'} z_{a'} . \quad (8.35)$$

- Ocular dominance patterns from a competitive Hebb rule



Feature-Based Models

- Considering multiple neuronal selectivities (e.g., retinotopic location (x, y) , ocular dominance (o) , orientation preference (e, θ)) simultaneously complicates the model.
- Instead of the selectivity of many neurons, we model the selectivity of just 5 "features" (in the example above).
- W_{ab} denotes the weight between output unit a and **feature** b .

$$z_a = \frac{\exp(-\sum_b (u_b - W_{ab})^2 / (2\sigma_b^2))}{\sum_{a'} \exp(-\sum_b (u_b - W_{a'b})^2 / (2\sigma_b^2))} . \quad (8.36)$$

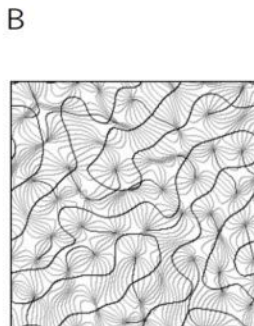
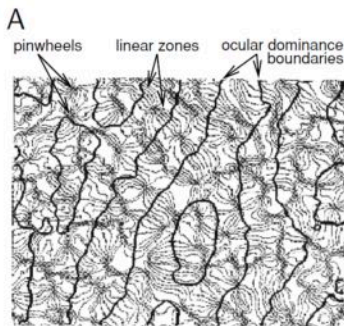
$$v_a = \sum_{a'} M_{aa'} z_{a'} . \quad (8.35)$$

- (8.36) introduces a competitive activity. We need some cooperative mechanism to ensure that nearby neurons have similar selectivities.
- One method is the self-organizing map (8.37), and another is the elastic net rule (8.38)
- Both rules make $W_{ab} \rightarrow u_b$ when v_a is positive.

$$\tau_w \frac{dW_{ab}}{dt} = \langle v_a (u_b - W_{ab}) \rangle. \quad (8.37)$$

$$\tau_w \frac{dW_{ab}}{dt} = \langle v_a (u_b - W_{ab}) \rangle + \beta \sum_{a' \text{ neighbor of } a} (W_{a'b} - W_{ab}), \quad (8.38)$$

- Figure A shows the arrangement of ocularity and orientation selectivity in the primary visual cortex of a macaque monkey.
- Figure B shows the result of an elastic net model. The similarity is "evident and striking"!



Anti-Hebbian Modification

- **The problem of redundancy** Without modifying $\mathbf{M}$ properly, Hebbian rules (e.g., Oja rule) set each row of the $\mathbf{W}$ to the principal eigenvector of $\mathbf{Q}$, making each output unit respond identically.
- To reduce the correlation between output units, anti-Hebbian modification causes recurrent interaction to decrease when there is simultaneous pre/postsynaptic activity.

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \alpha v^2 \mathbf{w}, \quad (8.16)$$

$$\tau_w \frac{dW_{ab}}{dt} = v_a u_b - \alpha v_a^2 W_{ab}, \quad (8.39)$$

$$\tau_M \frac{d\mathbf{M}}{dt} = -\mathbf{v}\mathbf{v} + \beta \mathbf{M} \quad \text{or} \quad \tau_M \frac{dM_{aa'}}{dt} = -v_a v_{a'} + \beta M_{aa'} \quad (8.40)$$

- There is a problem in (8.40) of weights decreasing to 0. Goodall(1960) proposed the following rule which pushes the weights towards the identity matrix.

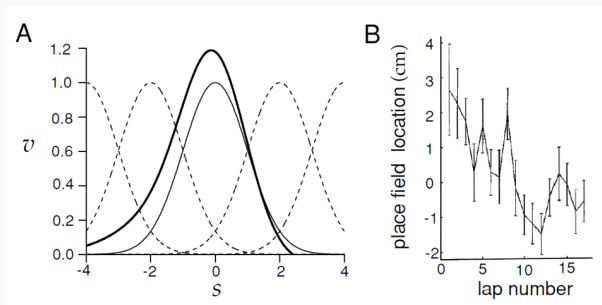
$$\tau_M \frac{d\mathbf{M}}{dt} = -(\mathbf{W} \cdot \mathbf{u})\mathbf{v} + \mathbf{I} - \mathbf{M}. \quad (8.41)$$

$$\langle (\mathbf{W} \cdot \mathbf{u})\mathbf{v} \rangle = \mathbf{I} - \mathbf{M} = \mathbf{K}^{-1}. \quad (8.42)$$

$$\langle \mathbf{v}\mathbf{v} \rangle = \langle (\mathbf{K} \cdot \mathbf{W} \cdot \mathbf{u})\mathbf{v} \rangle = \mathbf{I}. \quad (8.43)$$

Timing-Based Plasticity and Prediction

- Suppose $f_a(s)$ gives the pre-training average response of output unit a to stimulus s and attains its maximum at s_a .
- WLOG, let $s_1 = -2, s_2 = 0, s_3 = 2$. Suppose the stimulus increases with time (moves left to right in Figure A).
- The synapse from s_1 and s_2 will be strengthened (LTP) and the synapse from s_3 to s_2 will be weakened (LTD). The thick solid curve shows the new $f_2(s)$ which **predicts** the behavior of the stimulus.



Thank you for listening :D

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Supervised Learning Overview

- Perceptron learning rule
- Gradient descent learning with delta rule